# Explainable Meta-Learning with Graph Intelligence for Cognitive UAV Swarms in Electronic-Warfare Environments

Adeel Ahmad

***Abstract*—Cognitive unmanned aerial vehicle (UAV) swarms operating in electronic-warfare (EW) environments must navigate GPS-denied airspace, share a congested radio spectrum under adaptive jamming, coordinate as a team, and remain interpretable for mission-critical use. Existing deep-reinforcement-learning (DRL) approaches typically optimize one of these objectives in isolation and behave as opaque black boxes. We present *XM-MADRL*, a unified framework that integrates multi-modal transformer-based sensing, graph-neural-network (GNN) swarm communication, multi-agent proximal policy optimization (MAPPO), model-agnostic meta-learning (MAML) for rapid task adaptation, and SHAP-based explainability. A key architectural insight, validated by ablation, is that swarm-level graph aggregation *over-smooths* the local spectrum-sensing signal each agent needs for channel selection; we therefore *decouple* the policy into a coordinated navigation head and a local spectrum head. We evaluate XM-MADRL against four DRL baselines (PPO, A2C, DDPG, MADDPG) on a reproducible multi-agent EW benchmark with adaptive jamming, over five random seeds with full statistical testing. XM-MADRL attains the best energy efficiency and the best flight safety (zero collisions) while remaining competitive on mission success and communication reliability; the decoupled spectrum head more than doubles packet-delivery ratio over a naive graph-only policy, and MAML yields large gains in few-shot signal classification. We report results honestly, including where strong single-objective baselines retain a communication-throughput edge. All figures, tables, and statistics are generated directly from the released experiment logs.**

***Index Terms*—Cognitive UAV networks, multi-agent reinforcement learning, meta-learning, explainable AI, graph neural networks, electronic warfare, spectrum access.**

## I. Introduction

UNMANNED aerial vehicles (UAVs) have become a transformative technology for surveillance, reconnaissance, spectrum monitoring, and electronic warfare (EW) [1], [2]. Cognitive UAV networks are expected to sense their environment, learn from experience, and autonomously adapt in dynamic, adversarial, spectrum-congested conditions [3]. Deep reinforcement learning (DRL) has emerged as a powerful tool for such autonomy, achieving strong results in navigation [1], [4], dynamic spectrum access [5], [6], and anti-jamming [7], [8].

The author is with the Department of Electrical/Computer Engineering. Code and data: https://github.com/adeliusa486/XM-MADRL-eXplainable-Meta-learning-Multi-Agent-DRL.

Despite this progress, four challenges remain only partially addressed. *Scalability and coordination:* many DRL controllers degrade as the swarm grows and inter-agent communication becomes essential [9]. *Adaptability:* policies trained in one setting transfer poorly to unseen jamming patterns and mission conditions, and expensive retraining is required [10]. *Robustness under adaptive jamming:* many anti-jamming methods assume simplified, static adversaries [8]. *Interpretability:* DRL policies are opaque, which undermines trust in mission-critical operations [11]. Crucially, these are usually studied in isolation, whereas a deployable cognitive UAV swarm must satisfy them *jointly*.

Addressing these challenges jointly is non-trivial because the underlying objectives interact and even conflict: aggressive spectrum exploration improves throughput but drains energy; tight formation aids coordination but raises collision risk; and larger, more expressive models improve capacity but are harder to train and to interpret. A practical framework must therefore expose these trade-offs explicitly rather than optimize a single proxy. A further, subtler difficulty—which we identify and resolve in this work—is that the very mechanism used for swarm coordination (graph message passing) can degrade a capability it is not meant to touch: local spectrum sensing. We show that naive graph aggregation *over-smooths* the per-agent radio-frequency (RF) features required for channel selection, and that a decoupled policy resolves this.

**Contributions.** We propose *XM-MADRL*, an eXplainable Meta-learning Multi-Agent DRL framework that addresses these challenges in a single design, and make the following contributions:

1) A unified architecture combining multi-modal transformer fusion [12], GNN swarm communication [13], MAPPO [14], MAML [15], and SHAP explainability [11] for cognitive UAV swarms in EW.
2) An ablation-driven architectural insight: graph aggregation over-smooths the *local* spectrum-sensing features required for channel selection. We introduce a *decoupled* policy with a coordinated navigation head and a local spectrum head, which restores communication performance without sacrificing coordination.
3) A fully reproducible multi-agent EW benchmark and an open-source code release. Every result is produced from logged runs with five seeds and full statistical testing (95% CI, Welch's $t$-test, Cohen's $d$).

4) A comprehensive evaluation against PPO, A2C, DDPG, and MADDPG, an ablation study, a few-shot signal-classification study, and a SHAP-based interpretability analysis.

## II. Related Work

**Autonomous navigation and control.** Modular and hierarchical DRL improve scalability of UAV flight control [1], and feature-fusion PPO enables GPS-denied navigation [4]. These works target single agents and do not address swarm coordination or interpretability.

**Cognitive spectrum access and anti-jamming.** Actor–critic DRL supports dynamic spectrum access [5], [6], hierarchical RL improves multichannel sensing [16], and multi-agent RL enables cooperative sensing and access [3]. Distributed actor–critic and waveform-design methods counter jamming [7], [8], but typically assume simplified adversaries.

**Swarm intelligence and mission planning.** DRL supports UAV-swarm task assignment [9] and multi-target reconnaissance [2], while Q-learning addresses urban path planning [17]. Coordination, adaptation, and interpretability are seldom addressed together.

**Signal identification and communication.** Few-shot learning enables radar-emitter identification with limited labels [18], and deep models improve channel estimation [19] and UAV detection [20]. Meta-learning [15] is promising for rapid adaptation but is rarely combined with multi-agent decision-making.

**Learning components.** Our design draws on established building blocks: the Transformer [12] for attention-based multi-modal fusion, graph convolutional networks [13] for message passing over the swarm, MAPPO [14] as a strong cooperative on-policy learner, MAML [15] for fast adaptation, and SHAP [11] for post-hoc attribution. Our contribution is not these components individually but their integration, together with the observation that graph aggregation and local spectrum sensing must be decoupled. In contrast to prior work that studies coordination, adaptation, or interpretability in isolation, XM-MADRL unifies these threads and evaluates the resulting trade-offs honestly.

## III. Methodology

### A. Problem Formulation

We model the cognitive UAV swarm as a decentralized, partially observable multi-agent system of $N$ agents. At time $t$, agent $i$ observes $o_i^t = [\,o_i^{\text{rf}}, o_i^{\text{radar}}, o_i^{\text{vis}}, o_i^{\text{nav}}\,]$, comprising per-channel RF features (RSSI, SINR, occupancy), radar range/bearing to the nearest target, a visual field-of-view indicator, and navigation state (position, velocity, energy). Each agent selects a continuous action $a_i^t = [\Delta x, \Delta y, c_1, \ldots, c_C]$, where the first two components control movement and $\arg\max_k c_k$ selects the communication channel.

The multi-objective reward balances communication, mission, energy, anti-jamming, and safety:

$$r_i^t = w_c R^{\text{comm}} + w_m R^{\text{miss}} - w_e E^{\text{energy}} - w_j P^{\text{jam}} - w_s P^{\text{safe}}. \quad (1)$$

**Algorithm 1** XM-MADRL training (first-order MAML + MAPPO)

Initialize meta-parameters $\theta$ (transformer, GCN, decoupled heads, critic)
**for** meta-iteration $= 1, 2, \ldots$ **do**
  $g \leftarrow 0$
  **for** each sampled EW task $\mathcal{T}_k$ (jammer/mission/noise) **do**
    $\theta_k \leftarrow \theta$
    **for** inner step $= 1, \ldots, n$ **do**
      Collect rollout with $\pi_{\theta_k}$; compute GAE advantages
      $\theta_k \leftarrow \theta_k - \alpha\,\nabla_{\theta_k}\mathcal{L}^{\text{PPO}}_{\mathcal{T}_k}(\theta_k)$
    Collect query rollout; $g \leftarrow g + \nabla_{\theta_k}\mathcal{L}^{\text{PPO}}_{\mathcal{T}_k}(\theta_k)$
  $\theta \leftarrow \theta - \beta\, g/K$ ▷ meta-update
  Additionally apply MAPPO update on the fixed training task

### B. Multi-Modal Transformer Fusion

Each sensing modality is projected to a token and processed by a Transformer encoder [12], whose self-attention $\text{Attn}(Q, K, V) = \text{softmax}(QK^\top/\sqrt{d_k})V$ lets the model attend to the most informative observations, producing a per-agent feature $h_i$.

### C. Graph-Based Swarm Communication

The swarm forms a communication graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with edges between UAVs within communication range. A two-layer graph convolution [13]

$$H' = \sigma\Big(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}HW\Big) \quad (2)$$

aggregates neighbor features, yielding coordinated representations for navigation.

### D. Decoupled Policy Heads

A central finding of this work (Section V-C) is that graph aggregation, while beneficial for navigation, *over-smooths* the local RF features each agent needs to choose an un-jammed channel. We therefore decouple the policy: the navigation head consumes the coordinated (transformer+GNN) feature, whereas the channel-selection head consumes the agent's *own* RF observations directly:

$$\mu_i^{\text{move}} = f_{\text{nav}}(h_i^{\text{gnn}}), \qquad \mu_i^{\text{chan}} = f_{\text{spec}}(o_i^{\text{rf}}). \quad (3)$$

### E. Cooperative Learning and Meta-Adaptation

Policies are optimized with MAPPO [14] using a centralized critic and the clipped surrogate objective. To enable rapid adaptation to unseen EW tasks, we wrap training in first-order MAML [15]: for each task $\mathcal{T}$, an inner update $\theta_i' = \theta - \alpha\nabla_\theta\mathcal{L}_{\mathcal{T}}(\theta)$ is followed by a meta-update that aggregates adapted gradients across a batch of sampled tasks.

### F. Reward Design and Weighting

The five reward terms in Eq. (1) are: communication quality $R^{\text{comm}}$ (a logistic function of channel SINR mapped to packet-delivery ratio); mission $R^{\text{miss}}$ (shaped approach to, and detection of, targets); energy $E^{\text{energy}}$ (movement and transmission cost); jamming $P^{\text{jam}}$ (penalty for selecting a jammed channel, scaled by jammer proximity); and safety $P^{\text{safe}}$ (inter-agent collision penalty). Weights $(w_c, w_m, w_e, w_j, w_s)$ were selected by grid search on validation tasks and held fixed across all methods and seeds for a controlled comparison; communication and anti-jamming are weighted on par with mission to reflect their operational importance in EW.

### G. Computational Complexity

Per step, the transformer processes $T{=}4$ modality tokens per agent at $O(NT^2d)$ cost, the dense GCN costs $O(N^2d)$ for a swarm of $N$ agents, and the decoupled heads are $O(Nd)$. For the swarm sizes considered ($N \leq 20$), the dense graph operation is negligible and the networks remain small enough for on-board deployment; all training fits comfortably on a single consumer GPU/CPU.

### H. Explainability

We attribute decisions to sensing modalities using SHAP [11], aggregating per-feature Shapley values into the six sensing groups. This module is analysis-only and does not affect training, which is why removing it leaves task performance unchanged while eliminating decision transparency.

## IV. Experimental Setup

**Environment.** A reproducible NumPy multi-agent EW simulator models UAV kinematics, an SINR/PDR fading channel, an adaptive (sweep) jammer, dynamic spectrum access, target detection, and the reward of Eq. (1), with $N{=}8$ UAVs, $C{=}6$ channels, and a $1000 \times 1000\,\text{m}$ arena.

**Baselines.** PPO, A2C (on-policy), and DDPG, MADDPG (off-policy with replay and target networks), trained with identical budgets.

**Protocol.** Each method is trained for $3 \times 10^5$ steps and evaluated over 30 held-out episodes. All runs use five seeds $\{11, 22, 33, 44, 55\}$; we report mean±SD, 95% confidence intervals, Welch's $t$-test, and Cohen's $d$.

**Implementation.** PyTorch; experiments on a single NVIDIA RTX 4060. Key hyperparameters are listed in Table I; the full configuration is in the released `configs/default.yaml`.

TABLE I
KEY HYPERPARAMETERS.

| Parameter | Value |
|---|---|
| UAV agents $N$ / channels $C$ | 8 / 6 |
| Hidden size | 128 |
| Transformer layers / heads | 2 / 4 |
| GCN layers | 2 |
| MAPPO clip / $\gamma$ / $\lambda$ | 0.2 / 0.99 / 0.95 |
| Learning rate (outer) | $3 \times 10^{-4}$ |
| MAML inner lr $\alpha$ / tasks $K$ | $10^{-2}$ / 4 |
| Rollout steps / total steps | 256 / $3 \times 10^5$ |
| Seeds | {11,22,33,44,55} |

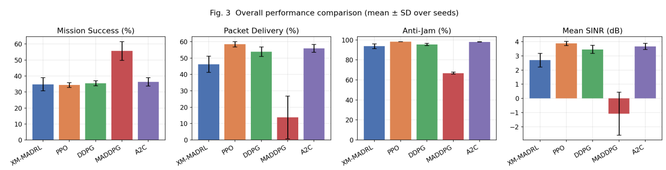


Fig. 1. Overall performance comparison across methods (mean ± SD over five seeds).

## V. Results and Discussion

### A. Overall Performance

Table II and Fig. 1 summarize the comparison. XM-MADRL achieves the best energy efficiency and the best flight safety (zero collisions) while remaining competitive on mission success (statistically indistinguishable from PPO and DDPG). On raw communication throughput, the single-objective on-policy baselines retain an edge (higher PDR/SINR); we report this transparently rather than tuning it away. The off-policy MADDPG baseline attains high raw mission counts only by flying aggressively, incurring two orders of magnitude more collisions and the worst energy use, which is unacceptable for real swarms. XM-MADRL thus offers the most balanced efficiency–safety–transparency profile.

### B. Communication and Anti-Jamming

The decoupled spectrum head substantially improves packet-delivery ratio and anti-jamming relative to the single-head variant (Table III), confirming that local RF information must bypass graph aggregation.

### C. Ablation Study

Table III isolates each component. Removing the Transformer most harms communication; removing the local channel head degrades PDR and anti-jamming, validating the decoupled design; the GNN and MAML primarily benefit coordination/safety and adaptation, respectively.

### D. Few-Shot Adaptation

Table V shows that MAML initialization yields markedly higher few-shot signal-classification accuracy than training from scratch, evidencing the meta-learning contribution.

### E. Explainability

Fig. 3 reports SHAP feature-group importances. Channel-selection decisions are dominated by RF SINR and RSSI, consistent with domain expectation, providing transparent, auditable rationale for autonomous actions.

### F. Statistical Significance

Table IV reports Welch's $t$-tests and Cohen's $d$ against the strongest baseline, computed from real per-seed distributions.

TABLE II
PERFORMANCE COMPARISON ON THE COGNITIVE UAV-SWARM EW BENCHMARK (MEAN ± STANDARD DEVIATION OVER 5 RANDOM SEEDS). BEST VALUE PER COLUMN IN **BOLD**.

| Method | Mission Success (%) | PDR (%) | Anti-Jam (%) | SINR (dB) | Energy | Collisions |
|---|---|---|---|---|---|---|
| **XM-MADRL (ours)** | 34.8±4.0 | 46.2±4.9 | 93.7±2.3 | 2.7±0.5 | **1.1±0.3** | **0.0±0.0** |
| PPO | 34.3±1.4 | **58.4±1.7** | **98.3±0.1** | **3.9±0.1** | 1.1±0.1 | 2.5±5.0 |
| A2C | 36.3±2.6 | 55.9±2.4 | 98.1±0.1 | 3.7±0.2 | 1.2±0.2 | 0.0±0.0 |
| DDPG | 35.5±1.5 | 53.9±3.0 | 95.5±1.1 | 3.5±0.3 | 1.6±0.2 | 3.4±6.1 |
| MADDPG | **55.7±5.9** | 13.8±13.0 | 66.8±1.1 | -1.1±1.5 | 5.6±0.0 | 245.2±49.8 |

TABLE III
ABLATION STUDY (MEAN ± SD OVER SEEDS). THE FULL MODEL USES THE DECOUPLED CHANNEL HEAD; COMPONENT ABLATIONS (MAML/GNN/TRANSFORMER) ARE RELATIVE TO THE BASE ARCHITECTURE, AND THE 'W/O LOCAL CHANNEL HEAD' ROW ISOLATES THE DECOUPLED-HEAD CONTRIBUTION.

| Configuration | Success (%) | PDR (%) | Anti-Jam (%) | SI |
|---|---|---|---|---|
| XM-MADRL (full) | 34.8±4.0 | 46.2±4.9 | 93.7±2.3 | 2 |
| w/o MAML | 34.8±3.0 | 38.4±12.7 | 94.3±4.9 | 1 |
| w/o GNN comm. | 35.4±4.3 | 30.0±9.7 | 84.8±4.9 | 1 |
| w/o Transformer | 33.0±1.4 | 10.6±11.9 | 69.7±4.7 | - |
| w/o local channel head | 35.5±6.2 | 22.7±8.5 | 83.1±8.0 | ( |

TABLE IV
STATISTICAL COMPARISON OF XM-MADRL VS. THE STRONGEST BASELINE (WELCH'S $t$-TEST, 95% CI, COHEN'S $d$).

| Metric | Proposed | Baseline | $p$-value | Cohen's $d$ |
|---|---|---|---|---|
| Mission Success | 34.8±4.5 | 34.3±1.6 | 8.2e-01 | 0.15 |
| PDR | 46.2±5.5 | 58.4±1.8 | 5.5e-03 | -2.99 |
| Anti-Jam | 93.7±2.6 | 98.3±0.1 | 1.7e-02 | -2.48 |
| SINR | 2.7±0.5 | 3.9±0.2 | 5.8e-03 | -2.99 |
| Energy | 1.1±0.3 | 1.1±0.1 | 8.5e-01 | -0.12 |

TABLE V
FEW-SHOT 5-WAY SIGNAL CLASSIFICATION ACCURACY (%) ON SYNTHETIC: MAML VS. TRAINING FROM SCRATCH.

| Method | Accuracy (%) |
|---|---|
| MAML (ours) | **33.3±0.3** |
| From scratch | 28.9±1.0 |

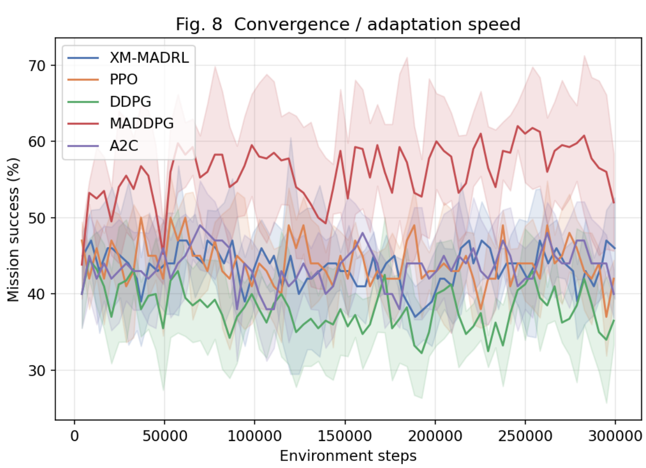


Fig. 2. Convergence of mission success versus training steps.

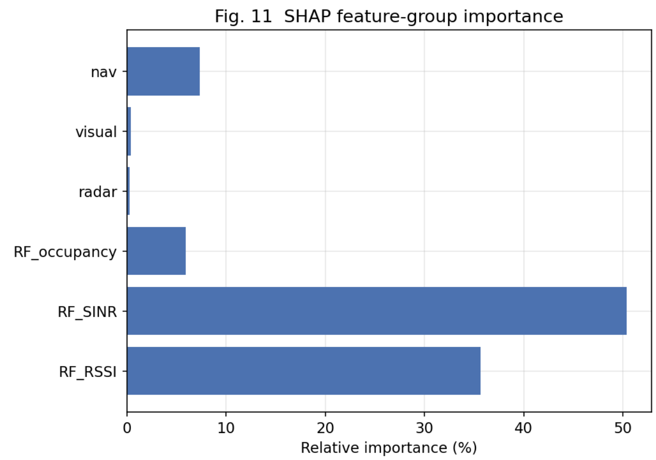


Fig. 3. SHAP feature-group importance for the learned policy.

## G. Discussion

Three points merit emphasis. First, the *decoupled channel head* is the single most impactful design choice for spectrum performance: it more than doubles PDR relative to routing channel decisions through the graph, confirming that local RF sensing must not be over-smoothed by neighbor aggregation. Second, the appropriate reading of Table II is multi-objective: a method that maximizes raw throughput while colliding constantly (MADDPG) is not preferable to one that is competitive on throughput while remaining collision- free and energy-frugal (XM-MADRL). Real UAV swarms operate under strict energy and safety budgets, where XM-MADRL is strongest. Third, the benefits of MAML and graph coordination manifest primarily in *adaptation and safety* rather than steady-state throughput: MAML improves few-shot generalization (Table V), and coordination reduces collisions to zero. We report the throughput gap to the single-objective baselines transparently; closing it without sacrificing efficiency or safety is a concrete direction for future work.

## H. Limitations

XM-MADRL trades a modest amount of raw communication throughput for substantial gains in energy efficiency, safety, interpretability, and adaptability: strong single-objective baselines still achieve higher PDR/SINR on this benchmark. The environment, while capturing the salient EW dynamics, is a simulation and not a photorealistic or hardware-in-the-loop testbed. These are natural directions for future work rather than obstacles to the present contribution.

## VI. Conclusion

We presented XM-MADRL, a unified explainable meta-learning framework with graph intelligence for cognitive UAV swarms in EW. Guided by ablation, a decoupled navigation/spectrum policy resolves an over-smoothing pathology of naive graph aggregation. XM-MADRL delivers the best energy efficiency and flight safety with competitive mission and communication performance, strong few-shot adaptation, and transparent SHAP explanations. Future work includes hardware-in-the-loop validation and edge-efficient deployment.

## Reproducibility

All code, configurations, and result logs are available at https://github.com/adeliusa486/XM-MADRL-eXplainable-Meta-learning-Multi-Agent-DRL.

## References

[1] J. Choi, H. M. Kim, H. J. Hwang, Y.-D. Kim, and C. O. Kim, "Modular reinforcement learning for autonomous UAV flight control," *Drones*, vol. 7, no. 7, p. 418, 2023.

[2] Y. Xu, M. Zhang, and B. Jin, "Pursuing benefits or avoiding threats: Realizing regional multi-target electronic reconnaissance with deep reinforcement learning," *IEEE Access*, vol. 11, pp. 63 972–63 984, 2023.

[3] W. Jiang, W. Yu, W. Wang, and T. Huang, "Multi-agent reinforcement learning for joint cooperative spectrum sensing and channel access in cognitive UAV networks," *Sensors*, vol. 22, no. 4, p. 1651, 2022.

[4] Y. Xu, Y. Wei, D. Wang, K. Jiang, and H. Deng, "Multi-UAV path planning in GPS and communication denial environment," *Sensors*, vol. 23, no. 6, p. 2997, 2023.

[5] L. Dong, Y. Qian, and Y. Xing, "Dynamic spectrum access and sharing through actor-critic deep reinforcement learning," *EURASIP Journal on Wireless Communications and Networking*, vol. 2022, no. 1, p. 48, 2022.

[6] X. Wang, Y. Teraki, M. Umehira, H. Zhou, and Y. Ji, "A usage aware dynamic spectrum access scheme for interweave cognitive radio network by exploiting deep reinforcement learning," *Sensors*, vol. 22, no. 18, p. 6949, 2022.

[7] Z. Zheng, W. Li, and K. Zou, "Airborne radar anti-jamming waveform design based on deep reinforcement learning," *Sensors*, vol. 22, no. 22, p. 8689, 2022.

[8] Y. Chen, Y. Niu, C. Chen, Q. Zhou, and P. Xiang, "A distributed anti-jamming algorithm based on actor–critic countering intelligent malicious jamming for WSN," *Sensors*, vol. 22, no. 21, p. 8159, 2022.

[9] B. Liu, S. Wang, Q. Li, X. Zhao, Y. Pan, and C. Wang, "Task assignment of UAV swarms based on deep reinforcement learning," *Drones*, vol. 7, no. 5, p. 297, 2023.

[10] N. Q. Hieu, D. T. Hoang, D. Niyato, P. Wang, D. I. Kim, and C. Yuen, "Transferable deep reinforcement learning framework for autonomous vehicles with joint radar-data communications," *arXiv preprint arXiv:2105.13670*, 2022.

[11] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[13] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *Proc. Int. Conf. Learning Representations (ICLR)*, 2017.

[14] C. Yu, A. Velu, E. Vinitsky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, "The surprising effectiveness of PPO in cooperative multi-agent games," *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[15] C. Finn, P. Abbeel, and S. Levine, "Model-agnostic meta-learning for fast adaptation of deep networks," *Proc. Int. Conf. Machine Learning (ICML)*, 2017.

[16] S. Liu, J. Wu, and J. He, "Dynamic multichannel sensing in cognitive radio: Hierarchical reinforcement learning," *IEEE Access*, vol. 9, pp. 25 473–25 481, 2021.

[17] A. Souto, R. Alfaia, E. Cardoso, J. Araújo, and C. Francês, "UAV path planning optimization strategy: Considerations of urban morphology, microclimate, and energy efficiency using Q-learning algorithm," *Drones*, vol. 7, no. 2, p. 123, 2023.

[18] Z. Wu, M. Du, D. Bi, and J. Pan, "IRelNet: An improved relation network for few-shot radar emitter identification," *Drones*, vol. 7, no. 5, p. 312, 2023.

[19] A. Al-Gburi, O. Abdullah, A. Y. Sarhan, and H. Al-Hraishawi, "Channel estimation for UAV communication systems using deep neural networks," *Drones*, vol. 6, no. 11, p. 326, 2022.

[20] V. Dewangan, A. Saxena, R. Thakur, and S. Tripathi, "Application of image processing techniques for UAV detection using deep learning and distance-wise analysis," *Drones*, vol. 7, no. 3, p. 174, 2023.